5.3 Factorising $z^{2n}+1$

WORKSHEET

The question

Show that $(z-\mathrm{e}^{\mathrm{i}\theta})(z-\mathrm{e}^{-\mathrm{i}\theta}) = z^2 - 2z\cos\theta + 1$.

Write down the $(2n)$th roots of -1 in the form $\mathrm{e}^{\mathrm{i}\theta}$, where $-\pi < \theta \leq \pi$, and deduce that

$$z^{2n}+1 = \prod_{k=1}^{n}\left(z^2 - 2z\cos\left(\tfrac{(2k-1)\pi}{2n}\right)+1\right).$$

Here, n is a positive integer, and the $\prod$ notation denotes the product.

(i) By substituting $z = \mathrm{i}$ show that, when n is even,

$$\cos\left(\tfrac{\pi}{2n}\right)\cos\left(\tfrac{3\pi}{2n}\right)\cos\left(\tfrac{5\pi}{2n}\right)\cdots\cos\left(\tfrac{(2n-1)\pi}{2n}\right) = (-1)^{\frac{1}{2}n}2^{1-n}.$$

(ii) Show that, when n is odd,

$$\cos^2\left(\tfrac{\pi}{2n}\right)\cos^2\left(\tfrac{3\pi}{2n}\right)\cos^2\left(\tfrac{5\pi}{2n}\right)\cdots\cos^2\left(\tfrac{(n-2)\pi}{2n}\right) = n2^{1-n}.$$

You may use without proof the fact that $1+z^{2n} = (1+z^2)(1-z^2+z^4-\cdots+z^{2n-2})$ when n is odd.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Where are the solutions of $z^{2n} = -1$ in the complex plane? Draw them for $n = 2$ and $n = 4$.

STAGE 3 Substituting $z = \mathrm{i}$

Ask yourself What is $\mathrm{i}^2 - 2\mathrm{i}\cos\theta + 1$? And what is $\mathrm{i}^{2n} + 1$ when n is even?

STAGE 4 When n is odd

Ask yourself For odd n, one of the angles θ_k equals $\frac{\pi}{2}$. Which one, and why does that make the substitution $z = \mathrm{i}$ useless as it stands?

Pause and try Divide both sides of the factorisation by $1 + z^2$, substitute $z = \mathrm{i}$, and simplify. You will need to pair up the remaining cosines.

Try this next

1 **A heptagonal product.** Use part (ii) with $n = 7$ to show that $\cos\frac{\pi}{14}\cos\frac{3\pi}{14}\cos\frac{5\pi}{14} = \frac{\sqrt{7}}{8}$. Then show that $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7} = \frac{1}{8}$. (*Hint:* relate the angles using $\cos\theta = \sin(\frac{\pi}{2} - \theta)$, or use the identity $\sin 2x = 2\sin x\cos x$ repeatedly.)

2 **Another substitution.** Substitute $z = 1$ in the factorisation to show that

$$\prod_{k=1}^{n} \sin\frac{(2k-1)\pi}{4n} = 2^{\frac{1}{2}-n}.$$

3 **Factorise $z^6 + 1$.** Write $z^6 + 1$ as a product of three real quadratics, and check your answer by multiplying out.

4 **Sines.** By factorising $z^{2n} - 1$ similarly and dividing by $z^2 - 1$, show that

$$\prod_{k=1}^{n-1} \sin\frac{k\pi}{2n} = \frac{\sqrt{n}}{2^{n-1}}.$$

Further reading

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 2 on roots of unity and polynomials, with beautiful pictures.

- **Benjamin Fine & Gerhard Rosenberger, *The Fundamental Theorem of Algebra*** (Springer). Several proofs of the theorem, from analysis, algebra and topology, written for undergraduates.
- **Paul Nahin, *An Imaginary Tale*** (Princeton University Press). A history of $\sqrt{-1}$, including roots of unity and their uses.

The full guided solution, and what the question leads to at university, is in Section 5.3:
https://beyondstep.keithmcnulty.org/ch05/z2n-plus-one.html